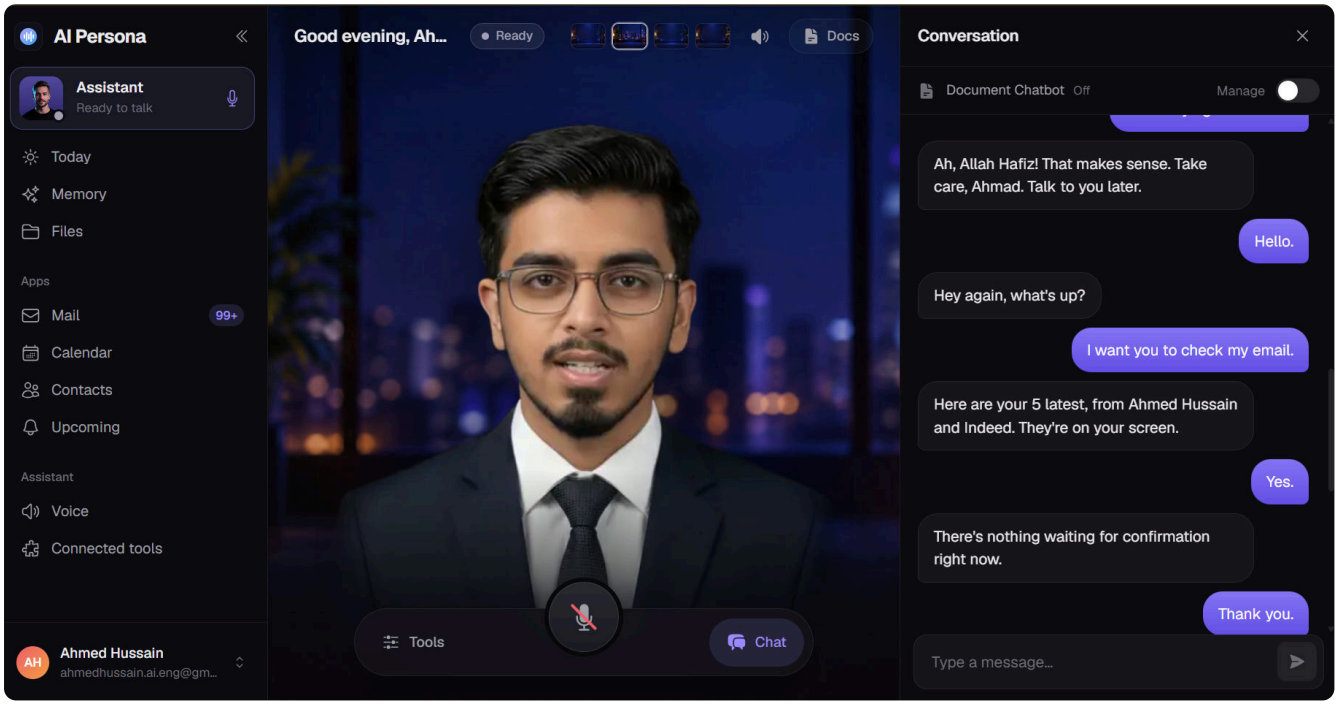
AI Persona

A real-time talking avatar that acts on your accounts, asks before it acts, and remembers you

Website · Demo video · Screenshots and material



Team	**Ahmed Hussain and Maria**
Institution	Department of Artificial Intelligence, COMSATS University Islamabad, Lahore Campus
Status	A working prototype, not a concept: an Android app, a web app and the full server system are built and running
Live at	https://aipersona.ahmedhussain.tech

At a glance

A face that speaks	Fast enough to talk to	Tested on a real account	Safe to trust
26.3 frames per second	**0.9 seconds**	**98% correct**	**Every change confirmed**
A talking avatar generated from the assistant's own speech on one GPU	From the end of the user's speech to the start of the reply (median)	On our 510-request benchmark, run end to end with real Gmail, Calendar, Drive and Contacts	Nothing is sent, booked or deleted without a yes

Contents: 1. Overview and Problem · 2. Solution · 3. Objectives and Outcomes · 4. Target Users · 5. Features and Deliverables · 6. Plan and Timeline · 7. Resources · 8. Risks · 9. Additional Details

1. Project Overview and Problem Statement

People run their day across apps that do not talk to each other: mail, calendar, contacts, files, a to-do list. Every small job means opening an app, tapping, and reading on a small screen.

Assistants should solve this. The ones people have today do not:

Today's option	What it does well	Where it falls short
Phone assistants (Siri, Google Assistant, Alexa)	Fast and always available	Act on personal data in only a few fixed ways, rarely confirm before acting, and guess or give up on a misheard name
Chat assistants (ChatGPT, Claude and similar)	Understand almost any request	Built for typing: replies take several seconds, they cannot be interrupted, and a tool call can invent a detail such as an email address
Voice-agent platforms	Let anyone build a talking agent	A voice alone is ordinary today, and many teams now build one
All of them	—	Do not remember the person well across conversations, and **none has a face**

> **The problem we address:** there is no assistant a person can both **see** and **trust with their own accounts**: one with a face that speaks in real time, fast enough for a real conversation, safe enough to send an email, and personal enough to know who "my supervisor" is.

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2. Proposed Solution

AI Persona is a personal assistant with a face. On the phone and in the browser, the user sees an avatar, talks to it like a person, and it answers in a natural voice while its lips, eyes and head move with the speech.

The avatar is the centre of the product, not decoration. It is generated from the assistant's own speech by a talking-head model, faster than real time on a single GPU, and it can be made from one photograph, so each user can have their own persona.

Behind the face is an agent that does real work. One spoken sentence is enough:

> "Email Maria and ask her for the slides by 5 PM. If she hasn't sent them by then, send a follow-up, and when she does, thank her."

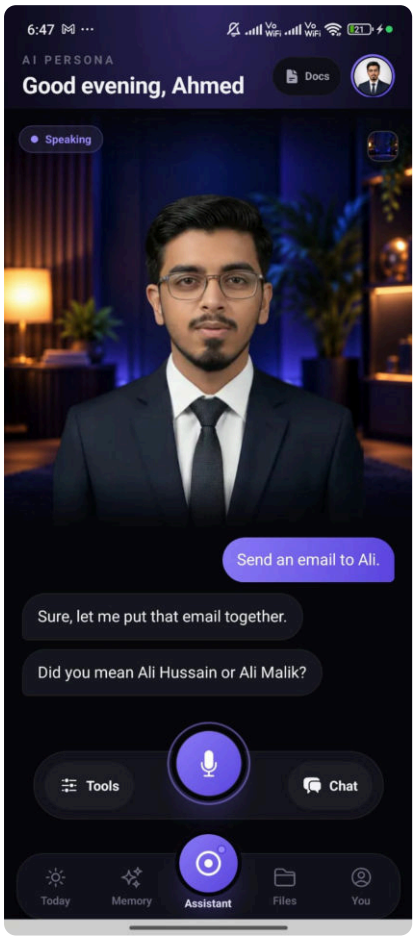
The assistant finds Maria in the user's contacts, writes the email, shows it on a card and waits for a "yes". It then watches the inbox and does the right thing later, even when the user is away, and tells them what happened.

From a spoken sentence to a finished task

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Speak  ->  Understand  ->  Plan  ->  Show and wait for "yes"  ->  Act  ->  Report back

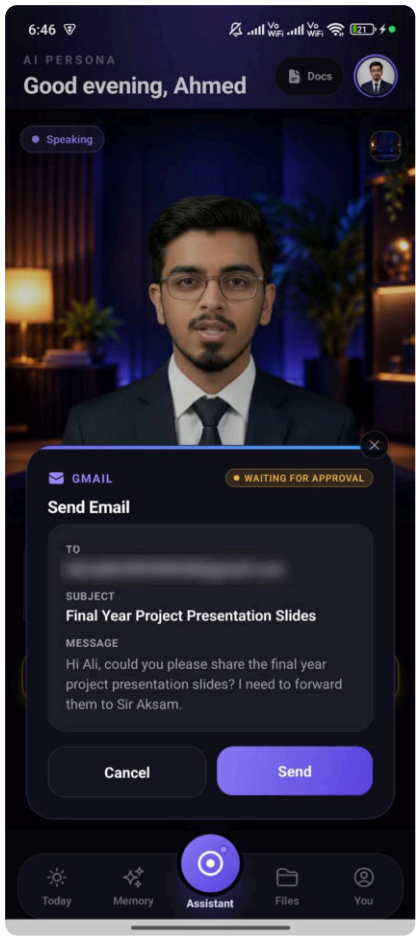
Every reply is spoken by the avatar, with its lips in step with the voice.
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What that looks like in the app



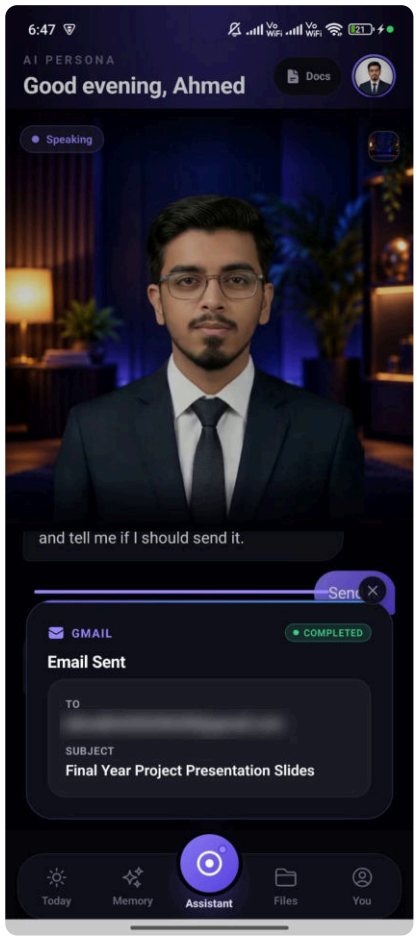
1. It asks, it never guesses

Two contacts are called Ali, so the avatar asks which one.



2. Nothing is sent without a yes

The finished email waits on a card while the avatar reads it out.



3. Done, and said so

After the yes, the card shows the email as sent.

How it is built

Part	Built with	What it does
Mobile app	React Native (Android)	Microphone, voice playback, cards, memory screen, avatar
Web app	React	The same assistant in a browser
Account backend	Node.js, MongoDB	Accounts, login, each user's connected tools
Agent engine	Python, FastAPI	The voice pipeline, tool choice, planning, confirmation, memory, speech
Avatar server	Ditto talking-head model on a GPU	Turns the assistant's speech into a talking face

> **One design rule runs through the whole system:** the language model decides what the user *means*; ordinary code decides what *happens*. The model reads the sentence, but code checks the name, builds the confirmation, and refuses anything the user did not approve.

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3. Objectives and Expected Outcomes

Objective	Target	Where we are
A real-time face	25 frames per second on one GPU, first frame within half a second	Achieved on the avatar server: 26.3 frames per second on one RTX 4090 in streaming mode, 258 ms of compute to the first frame. To be demonstrated live at the final
A reply fast enough to feel like conversation	The reply starts within 1 second of the user finishing	Measured on the live system: about 0.9 seconds (median) from the end of the user's speech to the start of the spoken reply, 0.77 seconds at its fastest
No change without permission	Every email, booking and deletion confirmed first	Built in: every action that changes something goes through a confirmation step
Never act on a misheard name	Ask instead of guess	Built in: "Did you mean Ali Hussain or Ali Malik?"
Remember the user	Facts kept across sessions, visible and removable by the user	Working: a memory graph the user can see and edit
Many users, separate data	Each user's own account, tools and memory	Working

Evidence so far: tested, not just built

What we tested	Result
The whole system, end to end. 510 requests on a live account with every tool real. A request passes only if every check on it holds: the right tool, the right details, the right reply	**98%** correct (90% on the first run, before fixing what it found)
New requests. 512 requests the system had never seen, on the same live account	**90%** correct
Memory. 50 questions from ten angles: working out an unstated relation, a wrong premise, counting, before and now, misheard names, Roman Urdu	**50 of 50** answered correctly
Reply speed on the live system	**0.9 seconds** (median) from the end of speech to the start of the reply
Avatar on one RTX 4090	**26.3 frames per second**, 1 late frame in 430
Automated tests of the engine	**850 tests**

Expected outcome: a persona a person can see and talk to every day, that runs their email, calendar, files and tasks, with nothing done behind their back.

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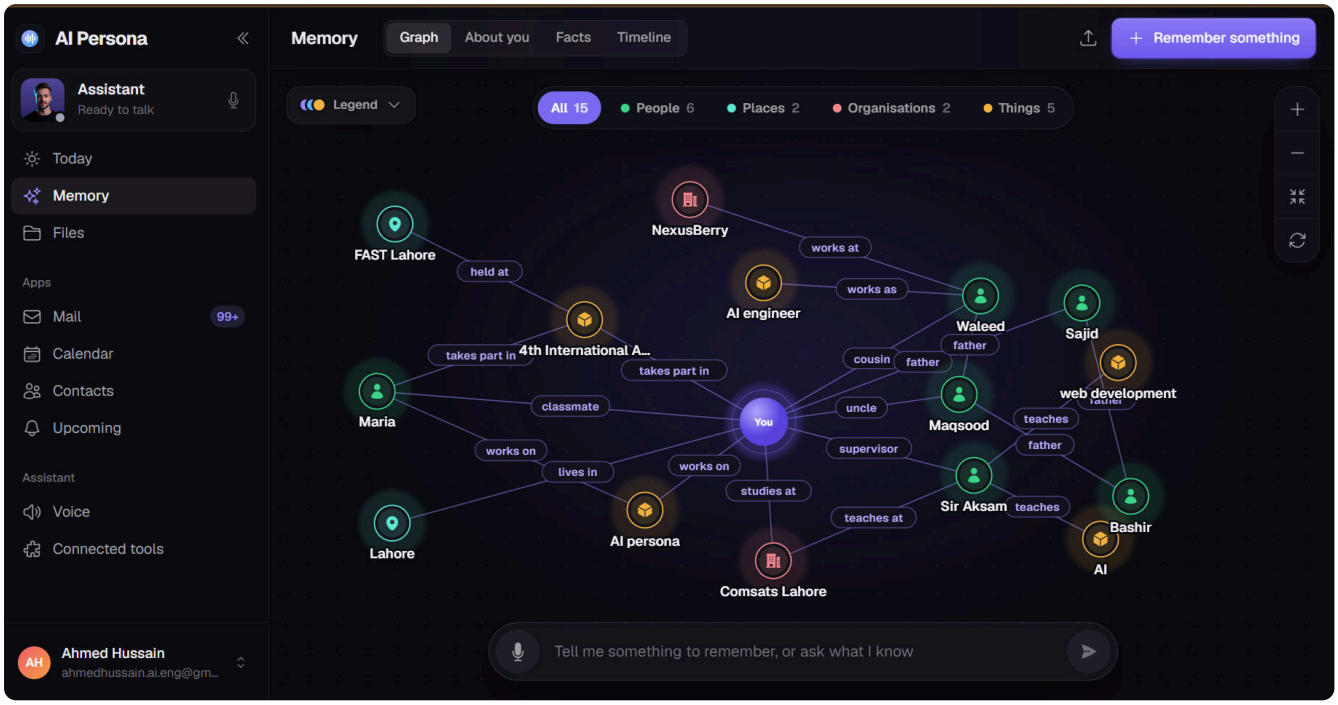
4. Target Users

Who	Why AI Persona fits
Students and professionals	Their work is spread over email, calendar and files
People whose hands or eyes are busy	Driving, cooking, walking: speaking is the only practical way to use a phone
People with low vision or difficulty typing	Voice and a face replace small text and small buttons
Speakers of Pakistani English and Roman Urdu	Built to understand broken English, Roman Urdu words and local names, with local touches such as prayer-time reminders for the user's city

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5. Key Features and Deliverables

Feature	What the user gets
A real-time talking avatar	A face generated from the assistant's speech, with lips, blinking and head movement, at video rate on one GPU. Made from a single photograph (a default persona or the user's own) and shown in a scene the user picks
Acts on the user's own accounts	Gmail (read, search, write, reply, send), Google Calendar (check, book, move, cancel; understands "the day after tomorrow", "next Friday", and "kal", Urdu for tomorrow), Contacts and Google Drive (found by meaning, not exact spelling), to-do list, reminders, weather, web search, nearby places
Answers from the user's documents	Upload a PDF and ask questions about it; the source is named
Works while the user is away	Conditional tasks in one sentence ("if she doesn't reply by 5, send a follow-up; if she does, thank her"), with a phone notification saying what was done
Safe by design	Nothing is sent, booked or deleted without a "yes". An unclear name is asked about, never guessed. Text inside an incoming email is treated as data, never as an instruction
Remembers, and the user is in control	Learns facts from conversation and links them in a graph, so it can answer "how is Waleed related to Sajid?" from what it was told. The user can see, correct or forget any of it, by tap or by voice
Feels like a conversation	The reply starts about 0.9 seconds after the user stops speaking, and the assistant can be interrupted at any moment; it tells real interruptions from "okay" and "yeah"
Open to other apps	Any app that offers an MCP server (the open standard for connecting tools to AI assistants) can be added as the user's own tool



The memory graph. What the assistant has learned about the user: people, places and things, with the relation written on each line.

Deliverables: the Android app, the web app, the server system with the avatar server, an evaluation report, and a live demo.

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6. Implementation Plan and Timeline

Period	Work	State
Dec 2025	Proposal and design	Done
Jan to Sep 2026	Real-time voice pipeline, the agent and its tools, confirmation flow, memory, documents, accounts for many users, the Android app, avatar model and benchmark	Done
Oct 2026	Scheduled tasks, memory graph, web app, the avatar in the app and on the web with selectable scenes, the live avatar demo	Shown live at the final on 24 October
Nov 2026	Fine-tune the avatar model on our own personas for sharper mouth detail and a steadier face; avatars from the user's own photo for every user	Planned
Dec 2026	Emotion: the avatar's tone and expression follow the conversation; Urdu speech alongside English	Planned
Jan 2027	Testing with outside users, reliability (backups, monitoring, alerts), first public release	Planned

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7. Resources and Requirements

Resource	What we use
Server	One small cloud server (2 CPU cores, 8 GB memory) runs the engine, the backend and the website today
GPU	One NVIDIA RTX 4090 (rented) for the live avatar. This is the main cost
Cloud services	Speech-to-text and text-to-speech (Deepgram), fast language models (Cerebras, with Groq and OpenAI as fallbacks), Google APIs for the user's own accounts, Firebase for notifications
Our own code	About 89,000 lines across four codebases (agent engine, mobile app, web app, account backend), with 12 tool agents
Free and open parts	Weather, prayer times, maps, web search and OCR all use free services or open-source software

8. Potential Challenges and Risks

Risk	What we do about it
Misheard names, especially Urdu names written in many spellings	Names are matched by meaning and sound, the user's own contacts are given to the recogniser as hints, and an unsure name becomes a question
The model inventing facts	Answers about the user come only from stored memory; a change is always shown and confirmed; "I don't know yet" is an allowed answer
GPU cost of a live avatar	One GPU serves the avatar faster than real time; a lighter pre-rendered avatar is the fallback when no GPU is available
Privacy of personal data	Each user's data is kept separate; Google access is the user's own and can be switched off per tool; memory is visible and removable
Dependence on outside services	Three language-model providers in a fallback chain; free services have a second source
Latency on weak networks	The pipeline overlaps its steps and streams speech in small pieces

9. Additional Details

Why this project

The championship looks for	What AI Persona brings
Innovation	A live talking avatar joined to an agent that really acts. Voice agents are common; a face that speaks in real time on top of a working, trusted agent is not. Its memory is a graph the user can see and edit, not a hidden profile
Technical depth	A real-time voice pipeline with a 0.9 second reply, a planner for requests with several parts, a confirmation step enforced in code, tasks that run later under a condition, a graph memory, and a talking-head model running faster than real time. All of it built, measured and tested
Real-world impact	It works today on real Gmail, Calendar, Drive and Contacts. It keeps working after the conversation ends and reports back. It is built for how people here actually speak: Pakistani English, Roman Urdu and local names

Where it can go next

The personal assistant is the first product, not the only one. A live avatar, a fast voice loop, an agent that uses tools safely and a memory of the person are what any talking persona needs: a teacher that remembers each student, a receptionist, a support agent. We are keeping the scope to the personal assistant until it is solid; these are directions, not promises.

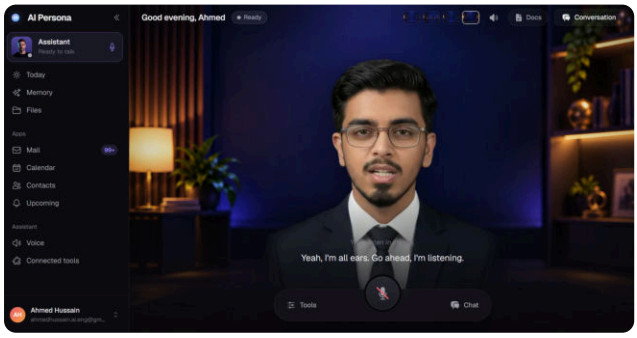
Demo video

A screen recording of the assistant in use: the avatar speaking, the memory graph, mail, calendar, contacts, voices and connected tools.

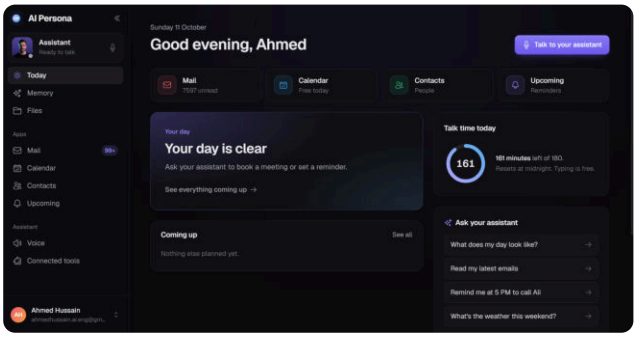
Watch the demo video

More screenshots

The web app



The avatar speaking, with its reply shown over the scene.

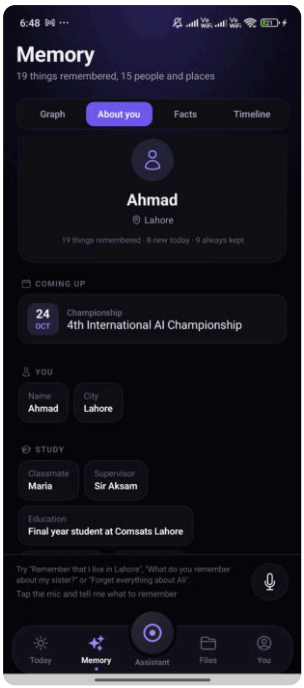


The Today page: mail, calendar, contacts and what is coming up.

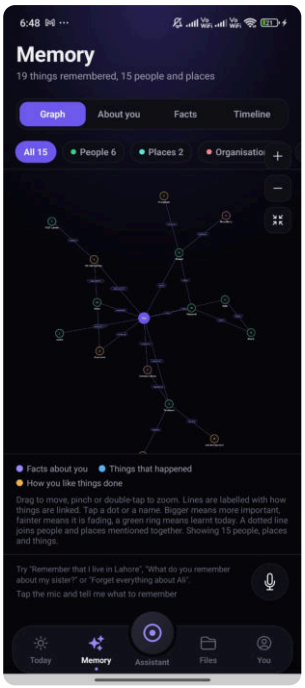
The mobile app



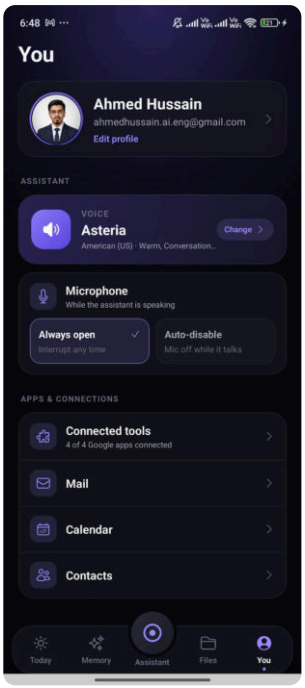
Answers with a card. The avatar speaks the weather and the card shows the figures.



What it knows about you. Facts sorted into groups by itself.



The memory graph on the phone.



The user's own assistant. Voice, microphone and connected apps.

AI Persona · Ahmed Hussain and Maria · COMSATS University Islamabad, Lahore Campus

Website · Demo video · GitHub